Automatic Classification of Palatal and Pharyngeal Wall Shape Categories from Speech Acoustics and Inverted Articulatory Signals

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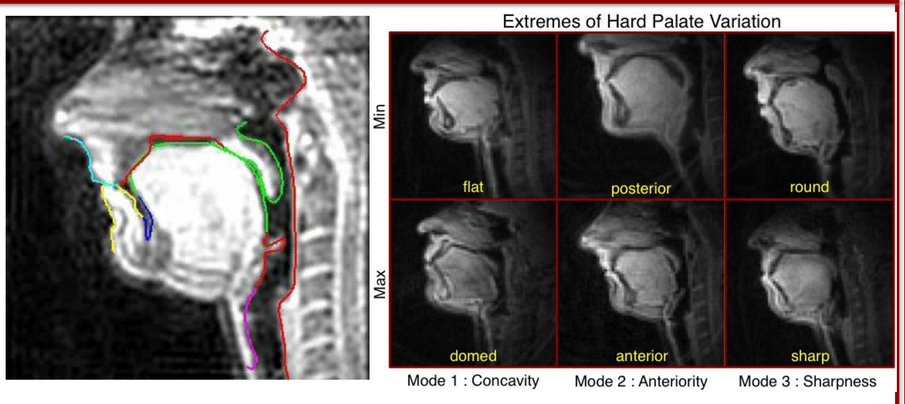
Outline

- Introduction
  - Aim to classify vocal tract morphological patterns
  - Using acoustics and inverted articulation together
- Methods
  - Open-smile features + GMM UWPP supervectors with SVM
  - Feature level fusion of MFCC + inverted articulatory features
- Database
  - In-house collected real time MRI data (rtMRI)
    - With accurate morphology measurement
    - Noisy speech
  - Wisconsin X-Ray Microbeam data (XRMB)
    - With morphology measurement (sensor placement may vary)
    - Clean speech
- Experimental results
  - 63% ~ 71% accuracy for concavity binary classification
  - Appending inverted articulatory features with MFCC helps

Morphological variability



- Vocal tract morphological variability
  - Vocal tract length [1,2,3,4]
  - Shapes of hard palate and rear pharyngeal wall [5,6]
  - Normalize for ASR (e.g. VTLN) or features for speaker ID

Articulatory variability

- Morphological variability also reflects on articulation
  - Flat palates exhibit less articulatory variability than highly domed palates during vowel production [7,8,9,10]
  - Articulation of coronal fricatives influenced by palate shape
    - apical vs. laminal articulation of sibilants [11]
    - jaw height and tongue body positioning [12,13].
  - Singing voice as an example

We believe that in order to make standard pronunciation, articulation need to compensate the vocal tract morphology.

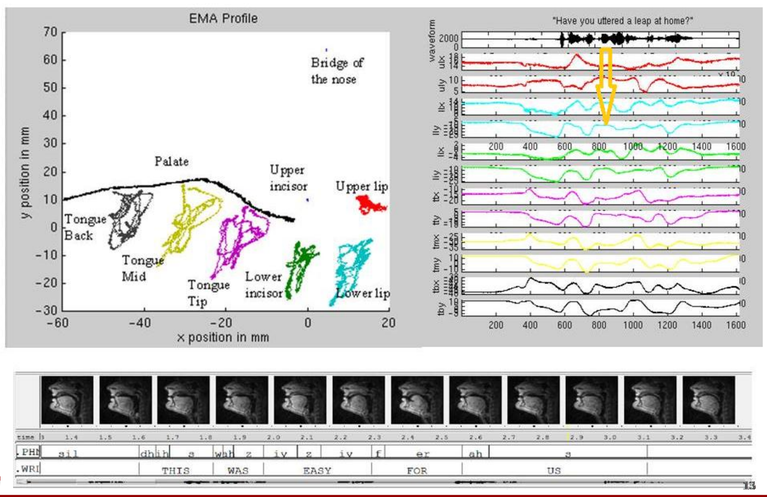
Different speakers articulate differently even for the same words

Acoustic-to-articulation inversion

For morphological categories classification

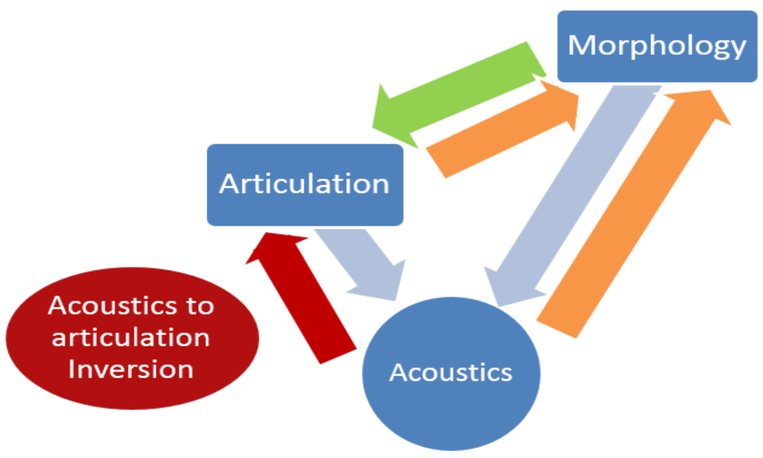
- Real articulation measurement impossible
- Speaker independent acoustic-to-articulatory inversion [14-15]
- Map inter-speaker acoustic variability to the reference speaker's intra-speaker articulatory variability
  - Female exemplar speaker in EMA, 460 MOCHA-TIMIT utterances

RAW MEASUREMENT FEATURES

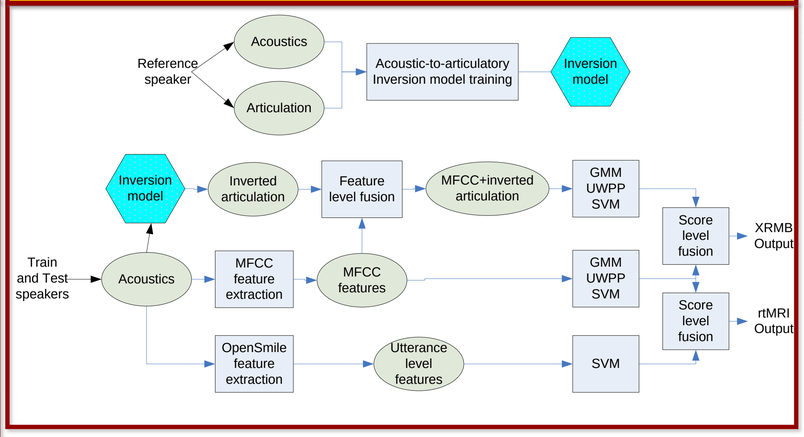


Morphology, articulation and acoustics

Acoustics + inverted articulation -> morphology

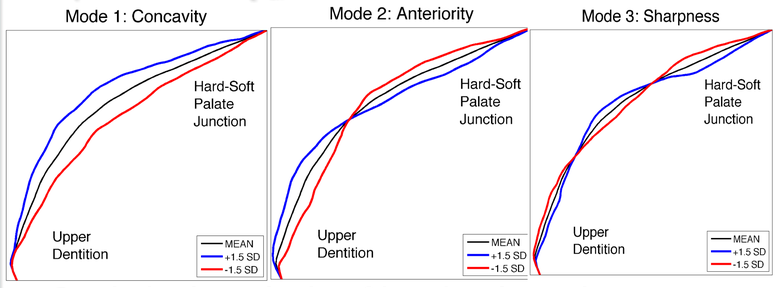


System Overview

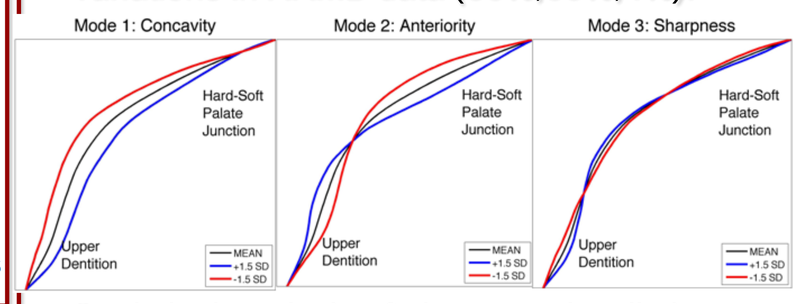


Morphological categories classification

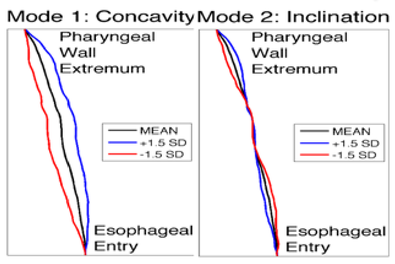
- Statistical analysis of hard palate shape variations in rtMRI data. (51%,25%,10%) (Figure reproduced from [6])



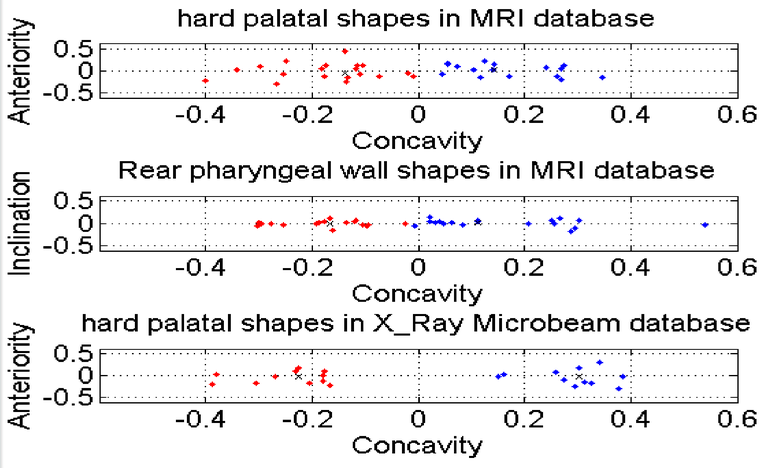
- Statistical analysis of hard palate shape variations in XRMB data (56%,35%,4%).



- Statistical analysis of pharyngeal wall shape variations in rtMRI data. (Figure reproduced from [6])



- K-mean clustering of morphology shapes cross all the speakers in each database



Databases, tasks and methods

- rtMRI database [6]

36 healthy speakers, age19-37 (mean 27, std 4.3)
22 American English, 8 German, 5 Mandarin, 1 Hindi
58 minutes total (1 min 37s per subject)

- XRMB data [16]

Sessions 1-101, JW11-63, 46 speakers, 4034 Utterances, average duration 5.72s per utterance
Concavity values between -0.15 and +0.15 removed
1893 utterances from 22 speakers

- Binary classification on the concavity

Limited data, moderate accuracy -> (no regression)
Apply K-mean clustering to get the binary categories
Concavity is the dominant for inter-speaker variability
Leave one speaker out cross validation

- Methods

Open-smile utterance level features [18]
UBM weight posterior probability supervector [19]
SVM classification with RBF kernel

Experimental results

- rtMRI database

	Hard palate			
Decision level	Utterance level		Speaker level	
Accuracy type	UA	WA	UA	WA
Open-smile + SVM	61	61	69	69
GMM MFCC UWPP + SVM	60	61	64	65
Open-smile + UWPP fusion			70	71
	Posterior pharyngeal wall			
Decision level	Utterance level		Speaker level	
Accuracy type	UA	WA	UA	WA
Open-smile + SVM	58	59	59	60
GMM MFCC UWPP + SVM	58	59	61	62
Open-smile + UWPP fusion			63	65

- XRMB database

	Hard palate	
Decision level	Speaker level	
Accuracy type	UA	WA
GMM MFCC UWPP + SVM	60	66
GMM (MFCC + inverted articulation) UWPP + SVM	63	69

Conclusions

- A feasible approach to automatically classify the palatal and pharyngeal wall shape categories from speech acoustics
- Adding inverted articulatory features helps
- Robust estimation of vocal tract morphology may be useful for handling this inter-speaker variability, for both ASR and speaker id applications.

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